

LESSON 3: CLOUDS & THINGS

The basic model for a computer was a device that provided the four basic functions of input, output, memory and processing. However, this model has gradually evolved, and the four basic functions decoupled and dispersed across a wide geographical area. No longer do these four basic functions have to occur on one device.

The internet brought about two advances in computing models, the cloud and the internet of things.

THE INTERNET BROUGHT ABOUT THE CLOUD AND THE INTERNET OF THINGS.

THE CLOUD

The term “cloud” refers to a system of servers that are accessed over the Internet. The term originated from the cloud-shaped symbol often used in diagrams to represent the internet.



The ability to connect to the internet and share information with other computers brought about possibilities. What if these computers could store a whole lot more of your information? What if these computers could provide additional computational power and could process information that could not be performed by your computer?

This additional storage and processing power would greatly enhance the functionality of your computer. Storage does not need to be on your device,

it can be anywhere as long as you can access it whenever you want and from wherever you want.

STORAGE CAN BE OFFERED AS A CLOUD-BASED SERVICE.

EXAMPLE 3.1

Mario likes taking photos. All the photos he takes are stored in the drive of his smartphone. But the more photos he takes the less space he has available. What will happen when one day the drive is full? In such a case, Mario needs more storage space. He has three options:

- 1. Memory Card:** If Mario's smartphone has a slot for a memory card, he could consider buying one to expand the storage. However, not all smartphones have this option, and even if his does, there's a limit to how much a memory card can store. Moreover, if Mario loses or damages his phone, he could potentially lose all his photos.
- 2. External Drive:** External drives like flash drives, hard disk drives (HDD), or solid state drives (SSD) are also viable options. They provide ample storage and can be a good choice if Mario prefers having physical control over his data. The downside is they can be lost or damaged. Additionally, accessing or sharing the data might be inconvenient as Mario will have to carry the external drive around.
- 3. Cloud-Based Storage Service:** Google Drive and Apple iCloud are examples of services that provide additional storage space for users and allow them to access files from any device that has internet access. They also automatically backup data, so if Mario loses his phone, his photos will still be safe. The downside is that if he doesn't have an internet connection, accessing his photos could be difficult. Also, storing data on the cloud means trusting the provider with the privacy and security of his data.

Storage Option	Advantages	Disadvantages	Maximum Storage Capacity	Approximate Price per GB
Memory Card (microSD card)	Portable and expandable storage. Immediate access to data.	Limited by phone compatibility and capacity. Risk of physical damage or loss.	Up to 1TB	Around \$0.20 - \$0.50
External Drive (HDD, SSD)	Large storage capacity. Data under physical control.	Risk of physical damage or loss. Not as portable or convenient as other options.	HDD: Up to 16TB SSD: Up to 8TB	HDD: Around \$0.02 - \$0.03, SSD: Around \$0.10 - \$0.20
Cloud Service (Google Drive, Apple iCloud)	Accessible from any device with internet. Automatic back-up and syncing.	Requires internet access. Recurring cost for extra storage. Dependence on service provider for data security.	For individual users, up to 2TB	Around \$0.025 - \$0.05

As for processing, that too can be provided as a cloud-based service. If you are connected to a much larger and more powerful computer then you could perform far more complex tasks than what your computer could perform alone.

PROCESSING CAN BE OFFERED AS A CLOUD-BASED SERVICE.

EXAMPLE 3.2

Liam loves playing video games. He has a basic laptop that he uses for school-work, but it's not powerful enough to run the latest high-definition games that require powerful processors and high-end graphics cards.

Google Stadia is a cloud-based gaming service. It allows Liam to play his favorite games directly from the cloud. The games are run on powerful servers,

and the video is streamed to Liam's laptop. His interactions, like button presses and mouse clicks, are sent back to the server allowing him to play more complex games than what he could on his computer.

A DATA CENTER IS A FACILITY THAT HOUSES SERVERS WORKING AS AN INTEGRATED SYSTEM SERVING THE REQUESTS OF MANY USERS.

DATA CENTERS

A datacenter is a facility that contains a great number of servers. The number of servers in a data center varies from tens to thousands depending on the size of the facility. It is important to realize that a data center is not simply a room filled with servers, it is a system of integrated components, such as servers, switches, power units, cooling systems and security systems, that come together to provide services such as the cloud-based services discussed in the previous section.

SERVERS

The heart of a data center is the server, a powerful computer designed to handle large amounts of data and run applications that serve other computers and their users across a network.

Servers are typically mounted on top of each other in metal frames called racks. Racks not only house the servers but also the other hardware components of the data center, such as the cables and switches that are used to connect servers together. In fact, a data center is one big, complex network consisting of servers connected by cables and switches.

Servers, like any other computer, have a basic architecture that also consists of the four basic components: input, output, memory and processing.

Although at certain terminals in a datacenter, there are keyboards, mice and

screens for data center administrators to take care of the internal workings of the servers, a server's main input and output component is a network adapter. This is because the majority of a server's input comes from the internet or the network on which it serves. Remember that the server is serving users by performing the tasks they request. As such, the server's input comes from the users and its output must be sent back to the users. So, whether it is input or output, the majority is being transmitted across a network, making the network adapter the most important I/O device for a server.

As for memory and processing components, servers also have central processing units (CPU), graphics processing units (GPU), caches, main memory and storage components.

PERFORMANCE

A data center is designed to respond to the requests of many users concurrently. Therefore, the performance of a server is measured by its throughput, which is defined as the number of tasks it can complete in 1 second.

A single server can handle thousands to tens of thousands of requests per second under regular conditions. Across an entire data center, the cumulative throughput could be in the millions or even billions of requests per second.

THROUGHPUT OF A SERVER IS THE NUMBER OF REQUESTS PROCESSED PER SECOND.

Throughput is dependent on the amount of data a server can receive in order to process, the more data it can process in one second the higher is its throughput. The amount of data that can be sent or received is referred to as the bandwidth which is measured as bits per second (bps), and ranges from megabits (Mbps) to gigabits (Gbps) and even to terabits (Tbps) per second.

Adequate bandwidth is critical to ensure that data can be transmitted rapidly,

enabling high-speed communication between servers and users or between different parts of the network.

Both throughput and bandwidth are interdependent. A server can have high throughput, but if the bandwidth is insufficient, data delivery to the server would be too slow, and the server won't be able to process tasks at its full capacity. On the other hand, a network with high bandwidth won't be fully utilized if the server's throughput is low.

BANDWIDTH IS THE AMOUNT OF BITS TRANSMITTED PER SECOND AND CAN LIMIT THE PERFORMANCE OF A DATA CENTER.

The performance of a server, its throughput, is enhanced by implementing parallel processing, which is the execution of multiple tasks at the same time. This method has already been implemented in the multi-core architecture of a CPU. Having multiple cores enhances the performance of a CPU by allowing it to process multiple tasks in parallel.

In the context of a data center, servers are arranged in clusters. This internal network organization allows the implementation of parallel processing at the level of the server. Multiple interconnected servers in a cluster can work simultaneously to perform multiple tasks for multiple users.

SERVERS ARE ARRANGED IN CLUSTERS THAT ALLOW THE PARALLEL PROCESSING OF TASKS.

Notes

Concurrent processing is processing by rapidly switching between tasks. Parallel processing is when processing of multiple tasks occurs simultaneously. Both enhance performance.

Data centers have two bandwidths, the internal and external bandwidths. This is because servers communicate with one another inside the data center via an internal network, and they also communicate with their clients via the network that connects them to the outside world.

DATA CENTERS HAVE BOTH AN INTERNAL AND EXTERNAL BANDWIDTHS.

If servers in a data center are working together then the internal bandwidth on the network interconnecting them must be high enough to not limit their performance. This is often quite high, with connections between servers and switches operating between 10 Gbps to 100 Gbps.

The external bandwidth of a data center, which is the total capacity for data transfer between the data center and the outside world, can also vary widely. Smaller data centers might have external bandwidth in the range of 1 to 10 Gbps, while large data centers might have external bandwidth in the order of Tbps.

From the perspective of the user, the most important measure of performance is the response time, the time it takes to receive a response. The response time factors in travel time through the internet to the data center, processing time at the data center, travel time back through the internet and any delays or wait time that occurs in between.

RESPONSE TIME IS THE TIME IT TAKES FOR A USER TO RECEIVE A RESPONSE FROM A SERVER TO A REQUEST.

DEPENDABILITY

Regardless of response time, throughput and bandwidth, users depend on data centers to run without any faults or failures. Users do not appreciate errors nor the failure of a server to respond. The dependability of a system such as a data center, refers to the extent to which it can operate without fault or failure.

DEPENDABILITY IS THE EXTENT TO WHICH A SYSTEM WILL OPERATE WITHOUT FAULT OR FAILURE.

Common reasons for faults or failures include over-heating, power outages, security breaches and disasters such as fire.

- 1. Over-heating:** Servers generate lots of heat due to the high volume of computational work they have to perform. Therefore, cooling systems such as Computer Room Air Conditioning (CRAC) units, Computer Room Air Handler (CRAH) units, and liquid cooling systems are essential to maintaining the proper operating temperature and preventing hardware failure.
- 2. Power outage:** Power disruptions can cause significant issues for data centers, including hardware damage and loss of data. Uninterrupted Power Supply (UPS) systems and backup generators are used to maintain power to the data center during outages.
- 3. Security Breach:** Physical and cyber security breaches can lead to data loss, service disruption, and significant harm to the data center's reputation. Physical and cyber security measures are implemented to protect the data center.
- 4. Disasters:** Natural disasters and accidents like fires or floods can cause catastrophic damage to data centers. Fire suppression systems, flood defenses, and disaster recovery protocols including off-site data backups and redundant systems are crucial for protecting the data center and ensuring uninterrupted service.

THE INTERNET OF THINGS

While the cloud has taken processing and storage capabilities beyond the confines of personal computers, the internet of things (IoT) extends input and output functionalities to a wide array of devices beyond traditional computers.

Computers are not the only devices connected to a network. A network may also connect other things. The term “things” refers to devices that can link to a network and either provide input or generate output forming an internet of things. IoT is a system of interrelated devices that are connected to a network and can send or receive data without the need for human interaction. These devices can include anything from small sensors to large appliances.

THE **IoT IS A SYSTEM OF INTERRELATED DEVICES CONNECTED TO A NETWORK AND TRANSMIT DATA.**

In the context of the IoT, a personal area network, PAN, can be created to connect a variety of smart devices to a central hub, often a smartphone or a personal computer.

EXAMPLE 3.3

Sam lives in a smart home. Using a central hub, a small computer designed for the purpose of smart home management, Sam can control and automate his home particularly because his home is full of smart things connected by a network to the hub.

Here are some of the smart things Sam has: smart watch, smart coffee machine, smart refrigerator, smart air conditioner, smart curtains, smart cameras and smart lights.

At 6:30 am, Sam’s smartphone alarm, synced with his sleep cycle via his smartwatch, wakes him up at the lightest point in his sleep, making waking

up less of a struggle. The moment he switches off the alarm, his smart coffee machine in the kitchen starts brewing his morning coffee.

After getting ready, Sam walks into the kitchen, where his smart refrigerator has already identified that he's running low on milk and eggs. The fridge automatically adds these items to Sam's grocery list on his smartphone, and it even suggests a healthy breakfast recipe based on the items currently available.

As Sam leaves his house for work, he doesn't need to worry about whether he has switched off all the lights or the AC. His smart home hub, which is connected to all the IoT devices in the house, knows when he's leaving due to his smartphone's GPS. It automatically turns off unnecessary lights and adjusts the thermostat to an energy-saving mode.

During the day, Sam's home security system keeps an eye on the house. The smart cameras and sensors monitor for any unexpected activity. If the cameras detect movement, Sam instantly receives a notification on his phone. He can watch a live feed and, if necessary, sound an alarm or contact the authorities, all from his phone.

Once Sam heads back home, his smart home hub detects his smartphone's location and begins to prepare for his arrival. The thermostat adjusts to a comfortable temperature, and the lights turn on as he walks in. As he steps into the house, his smart home hub greets him and briefs him about his schedule for the next day, the weather forecast, and any unread emails or messages.

At night, Sam can use his voice to control his home. A simple command like, "Good night, Home," can cause the lights to dim, the doors to lock, the alarm system to activate, and his favorite relaxing playlist to start playing softly in the background.

In this manner, each device in Sam's smart home plays a role in automating routine tasks, enhancing his comfort, saving energy, and providing him with peace of mind about his home's security. All these devices form a network around Sam, working together to make his life easier and more efficient.

An IoT is not restricted by distance, as long as there is network coverage. Various sensors and smart devices can connect to a central computer. Areas as wide as farms and cities can be covered by IoT. Here are two examples.

THE **IoT** IS NOT RESTRICTED BY DISTANCE AS LONG AS THERE IS NETWORK COVERAGE.

EXAMPLE 3.4

Meet Farmer Maya, an innovative agriculturist using smart farming to optimize her operations. At dawn, Maya's weather station, equipped with sensors, provides a precise forecast, helping her plan the day's tasks. Her smart irrigation system, connected to soil moisture sensors, already knows which parts of her farm need water. These sensors feed data to the system, which autonomously controls the water flow, ensuring each crop gets the right amount of hydration while conserving water.

Maya has drones equipped with infrared cameras. When flown over her fields, they provide valuable data about crop health, identifying potential disease outbreaks or pest infestations before they become visible to the human eye.

Livestock wear connected devices that monitor their health and location. If a cow's activity level drops or it strays from the herd, Maya gets a notification on her phone.

At the end of the day, all data collected is sent to Maya's computer. Thanks to her smart farm, Maya ensures that her crops grow well, her livestock are healthy and that she uses her water resources efficiently.

EXAMPLE 3.5

Director Dave, is in charge of a city's waste management department and has implemented smart technology to optimize garbage collection.

To start the day, Dave receives an automated report generated by smart sensors installed in public garbage bins throughout the city. These sensors monitor the fill levels of the bins and report back when they're nearing capacity. This real-time data helps Dave's team plan their collection routes, ensuring they only visit bins that need to be emptied and reducing unnecessary trips.

Dave's garbage trucks are equipped with GPS and route optimization software. This allows the central system to direct each truck along the most efficient route, saving time and reducing fuel consumption.

The smart waste bins also have sensors that identify different types of waste for recycling. This helps the city to improve its recycling rate and reduces the volume of waste sent to landfills.

With this smart waste management system, Dave has made garbage collection more efficient, improved recycling rates, and made his city cleaner and more sustainable.

Career Tips

IoT Solution Architects: These individuals design comprehensive IoT solutions, ensuring various components like devices, cloud services, and connectivity solutions work seamlessly together.

IoT Hardware Engineers: They are responsible for designing and developing the hardware aspects of IoT devices, such as sensors, circuits, and embedded systems.

SUMMARY

The internet brought about the cloud and the internet of things.

The cloud is a system of servers that can be accessed over the internet. The cloud offers additional storage and processing services.

Additional storage can be obtained by using memory cards, external drives and cloud-based storage services. The advantages of cloud-based storage are its accessibility from any device with internet access and automatic backups ensure the most recent information is kept secure. The disadvantages are its dependence on the internet and service providers for accessibility and security, as well as its potential recurring cost.

A data center is a facility that houses servers working as an integrated system serving the requests of many users. It is a system of integrated components, such as servers, switches, power units, cooling systems and security systems.

The server is the heart of a data center. It is a powerful computer designed to handle large amounts of data and run applications that serve other computers and their users across a network.

Throughput is a measure of server performance and is defined as the number of requests processed per second. Throughput is dependent on bandwidth which is the amount of data that can be sent or received. Data centers have both an internal and external bandwidths. Servers communicate with one another inside the data center and they also communicate with clients via an external network.

From the perspective of the user, the most important measure of performance is the response time, the time it takes to receive a response.

The performance of a server is enhanced by parallel processing. Servers are arranged in clusters, which allows the for parallel processing. Servers in a cluster work simultaneously to perform multiple tasks for multiple users.

Dependability is the extent to which a system will operate without fault or failure, which can be caused by over-heating, power outages, security breaches and disasters such as fire.

The IoT is a system of interrelated devices connected to a network and transmit data autonomously. The IoT extends input and output functionalities to a wide array of devices beyond traditional computers. All these devices are connected to a central hub.

The IoT made smart homes, smart farms and smart cities possible.

Activity 3.1

Read the following statements, and write on the line provided for each statement 'T' if it is true and 'F' if it is false.

T/F	STATEMENT
<u>F</u>	The cloud does not depend on the internet.
<u>T</u>	Both storage and processing can be provided as cloud-based services.
<u>T</u>	A data center is a facility with an integrated system of servers.
<u>T</u>	Servers, like other computers, have the four basic components of input, output, memory and processing.
<u>F</u>	Throughput is a measure of dependability.
<u>T</u>	Bandwidth refers to the amount of data transmitted per second.
<u>T</u>	To improve performance servers are arranged in clusters that perform parallel processing.
<u>T</u>	Dependability is the extent to which a system will operate without fault or failure.
<u>T</u>	The IoT is a system of interrelated devices that are connected and can transmit data.
<u>F</u>	The IoT only includes devices in close proximity.

Activity 3.2

Answer the questions below.

1. What is the “cloud” in cloud computing?
 - a) A weather system
 - b) A system of servers accessed over the Internet
 - c) A data storage device

2. Which of the following is not an advantage of cloud storage?
 - a) Accessible from any device with internet
 - b) Automatic backup and syncing
 - c) Doesn't require an internet connection

3. Which of the following tasks can be performed by a cloud-based service?
 - a) Only storage
 - b) Only processing
 - c) Both storage and processing

4. Which additional storage option provides the greatest space?
 - a) cloud-based storage
 - b) memory card
 - c) external drive

5. Which additional storage option depends on the internet?
 - a) cloud-based storage
 - b) memory card
 - c) external drive

Activity 3.3

Answer the questions below.

1. What is the main component of a data center?
 - a) Cooling system
 - b) Network switch
 - c) **Server**

2. What does the term “throughput” refer to in the context of a server?
 - a) The amount of data a server can store
 - b) **The number of tasks a server can complete in 1 second**
 - c) The speed at which a server can download data

3. What is “bandwidth” in the context of data center performance?
 - a) The physical width of the cables connecting servers
 - b) The number of servers in a data center
 - c) **The amount of data that can be sent or received per second**

4. What is the measure of performance from the user’s perspective?
 - a) Throughput
 - b) Bandwidth
 - c) **Response time**

5. What does “dependability” in the context of a data center refer to?
 - a) The amount of data a server can store
 - b) **The extent to which a system will operate without fault or failure**
 - c) The speed at which a server can download data

Activity 3.4

Answer the questions below.

1. What does the term “Internet of Things” refer to?
 - a) The network connecting all the computers in the world
 - b) A network of interrelated devices that can transmit data without the need for human interaction
 - c) A network of satellites providing global internet coverage
2. What can smart home devices do in the context of IoT?
 - a) Automate routine tasks
 - b) Enhance comfort
 - c) Both a and b
3. What is a central hub in the context of IoT?
 - a) A device that connects to other devices on a network and manages them
 - b) The main server in a data center
 - c) The main router in a network
4. How does smart farming help in conserving water?
 - a) By watering all crops equally irrespective of their needs
 - b) By using soil moisture sensors to autonomously control the water flow
 - c) By only watering the crops during the night
5. How can a city’s waste management department benefit from IoT?
 - a) By having to manually check the fill levels of the bins
 - b) By planning their collection routes based on real-time data about the fill levels of the bins
 - c) By increasing the volume of waste sent to landfills

Activity 3.5

Answer the questions below.

1. How has the internet contributed to the evolution of the basic model for a computer?

The internet has allowed for the decoupling and geographical dispersal of the four basic computer functions: input, output, memory, and processing. This has given rise to new computing models such as cloud computing and the Internet of Things.

2. What are some potential advantages and disadvantages of using a cloud-based storage service?

Advantages of cloud-based storage services include accessibility from any device with internet, automatic backup and syncing of data. Disadvantages include the requirement of an internet connection to access data, potential security and privacy concerns, and the recurring cost for extra storage.

3. How can cloud-based services enhance the functionality of your computer?

Cloud-based services can provide additional storage and processing power beyond the capabilities of one's personal computer. This allows for more complex tasks to be performed and larger amounts of data to be stored than what could be achieved with the computer alone.

Activity 3.6

Answer the questions below.

1. What are the main components of a data center?

The main components of a data center are servers, switches, power units, cooling systems and security systems.

2. How can bandwidth limitations impact the performance of a data center?

Bandwidth limitations can limit the amount of data that can be sent to and processed by the servers, reducing their throughput. If the bandwidth is insufficient, data delivery to the servers would be too slow, and the servers won't be able to process tasks at their full capacity.

3. What measures can be taken to ensure the dependability of a data center?

Measures to ensure the dependability of a data center include cooling systems to prevent overheating, Uninterrupted Power Supply systems and backup generators to maintain power during outages, physical and cyber security measures to protect against security breaches, and fire suppression systems,

Activity 3.7

Answer the questions below.

1. How do smart home devices improve a user's quality of life?

Smart home devices improve a user's quality of life by automating routine tasks, enhancing comfort, providing security, and increasing efficiency. They can adjust to the user's habits and preferences, such as brewing coffee when the alarm goes off, or suggesting recipes based on available ingredients.

2. What is one way smart farming benefits the environment?

Smart farming benefits the environment by optimizing water usage, with soil moisture sensors controlling the water flow to ensure each crop gets the right amount of hydration. This helps in conserving water.

3. How does a smart waste management system contribute to a cleaner and more sustainable city?

A smart waste management system contributes to a cleaner and more sustainable city by optimizing garbage collection, reducing unnecessary trips, and saving time and fuel. Sensors in the bins monitor fill levels and enable collection only when necessary. They also identify different types of waste for recycling, improving recycling rates and reducing the volume of waste sent to landfills.



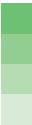
Activity 3.8

Cloud Storage Activity: Sign up for a free cloud storage service such as Google Drive or Dropbox. Upload a document or picture from your computer, and then access and download it using a different device. This will demonstrate the concepts of cloud storage and accessibility from multiple devices.



Activity 3.9

Exploring Data Center Virtual Tour: Many data centers offer virtual tours online (like Google data centers). Explore these tours to understand the scale, organization, and components of a data center, including servers, cooling systems, and security measures. Then discuss or write a report on your findings.



Activity 3.10

IoT Smart Home Design: Divide students into groups and ask each group to design a smart home using IoT devices. Students can research different types of IoT devices, decide what their smart home would include, and explain how the devices would communicate with each other and the central hub. This project can be presented as a diagram, a report, or a presentation.



Activity 3.11

Creating a Smart City Map: Divide students into groups and ask each group to design a “smart city” that utilizes IoT technology. Students can research different applications of IoT in urban environments (like smart traffic lights, waste management systems, air quality sensors) and incorporate these into their city. They can present their city map and explain how each IoT element contributes to the efficiency, sustainability, or livability of the city.

